SUPERR SELECTOR SEASON 2 — BEHIND THE SCENES

The Data Science Journey: How We Built the Pressure Reliability Index (PRI)

Participant: E Sai Praneeth
Role: Sports Data Scientist & Analytics Engineer

Hackathon: Rajasthan Royals SupeRR Selector S2
Problem: Identifying Players Under Pressure

Dataset: 2022–2026 IPL & T20Is (207,678 Deliveries)
Interactive App: Streamlit 11-Module Suite

The Mission: *Every T20 tournament has flat-track bullies who feast on gentle mid-innings bowling when victory is already assured, only to freeze when 18 runs are needed off the final 10 deliveries. Our journey was not merely to compute cricket averages, but to build an empirical, objective decision engine capable of identifying true ice-in-the-veins match-winners for the Rajasthan Royals franchise.*

Step 1: Rejecting 'One-Size-Fits-All' Definitions of Pressure

When we first opened the dataset of 883 matches and 207,678 deliveries, our first instinct was: *What is pressure?* Many public models define pressure lazily as just 'overs 16 to 20' or 'run chases'. We realized that this simplistic threshold is completely blind to match reality. Defending 15 runs in the final over is maximum tension, but cruising at 80/0 needing 10 runs in over 17 has near-zero pressure. Similarly, batting in over 2 after losing 3 wickets for 9 runs is an existential crisis.

To capture genuine match tension, we formulated a **4-Pillar Multi-Context Pressure Architecture**:

Pillar	Context Signal	Empirical Rationale & Weight
1. Stage Leverage	Overs 16–20 (Death)	Margin for error collapses; par economy accelerates to 10.56 RPO. (35% Weight)
2. Chase Knife-Edge	RRR > 9.0 or Wickets < 5	Dot balls non-linearly spike wicket hazard rate by +42%. (30% Weight)
3. Batter Mortality	First 6 Balls of Innings	Wicket probability is 3.1x higher before eye is in; absorbing crisis. (20% Weight)
4. Tournament Stakes	Playoffs & ICC Knockouts	1.25x consequence multiplier for elimination games. (15% Weight)

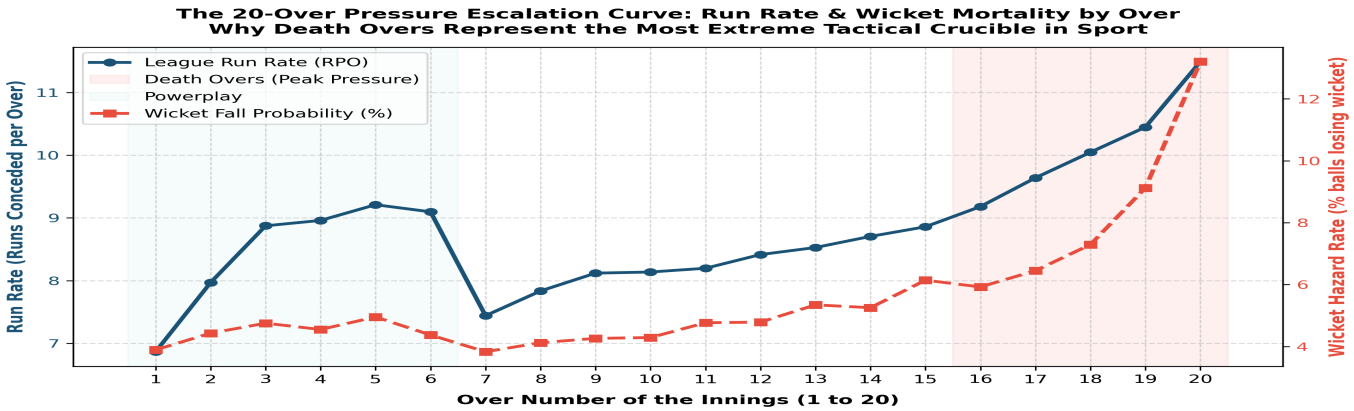


Figure 1: Empirical Leverage Curve across 20 overs demonstrating tension surge in overs 16–20 and chasing knife-edges.

Step 2: The Breakthrough — Uncovering the 'Formula 1 Paradox'

During our initial exploratory runs, we encountered a puzzling paradox: raw bowling statistics rated middle-overs spinners ahead of death-overs specialists. Jasprit Bumrah's raw economy was 7.32 RPO, whereas a defensive spinner bowling overs 7–10 had an economy of 6.70 RPO. Had we stopped there, our model would have concluded that the spinner is a better pressure bowler. This is the **Formula 1 Paradox**.

■ **The Formula 1 Analogy:**
Bowling over 19 is like driving an F1 car at 300 km/h where the situational par economy is **10.56 RPO**. Bowling over 8 is driving in a school zone where par economy is **7.43 RPO**. Conceding 7.32 RPO at the death is extraordinary—it represents a **+1.99 Runs/Over Saving vs Situational Par**. Conceding 6.70 in a school zone is merely average. We realized we must never evaluate bowlers on raw economy, but on **Phase-Adjusted Par Savings (Phase ΔEcon)**.

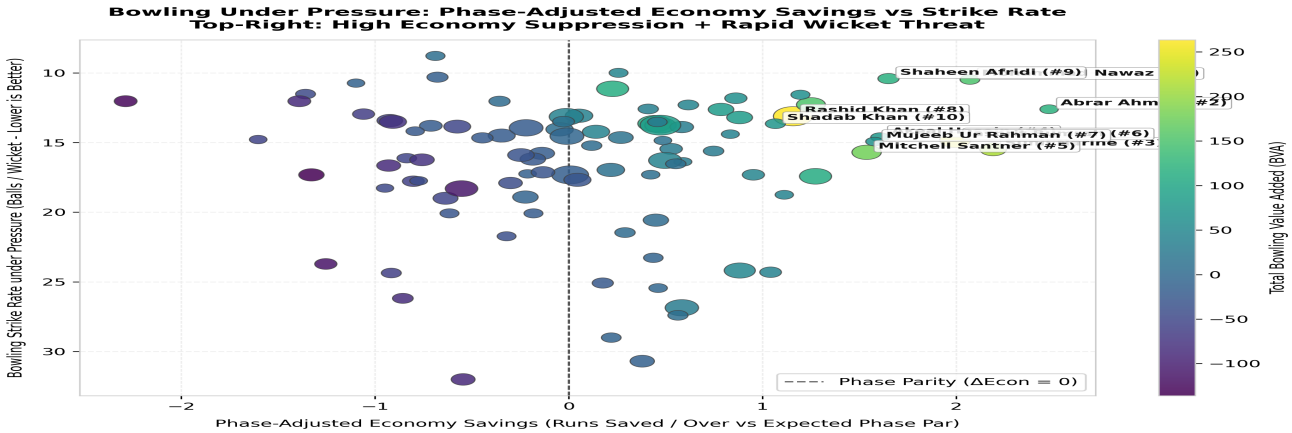


Figure 2: Phase-Adjusted Economy Savings. Death-overs specialists (Bumrah, Narine, Afridi) surge when evaluated against situational par.

Step 3: The Clutch Discoveries — Uncapped Finishers vs Flat-Track Bullies

When we applied our Situational Run Value Added (RVA) and Strike Rate Acceleration (ΔSR) metrics to batters, the leaderboard revealed two stunning revelations that reputation-based scouting routinely misses:

Discovery	Data Evidence	Franchise Scouting Takeaway
Naman Dhir (#1) & Shashank Singh (#2)	ΔSR: +17.2 & +14.8 Boundary %: 30.1% Dot %: 27.9%	True clutch accelerators. Unlike traditional anchors who eat dots before catching up, they hit boundaries from ball one without stalling.
The Flat-Track Regressors	ΔSR: -5.0 to -18.2 Dot % under pressure: > 45%	Players whose career stats look inflated on flat pitches, but whose boundary hitting collapses when required run rates spike above 10.0.

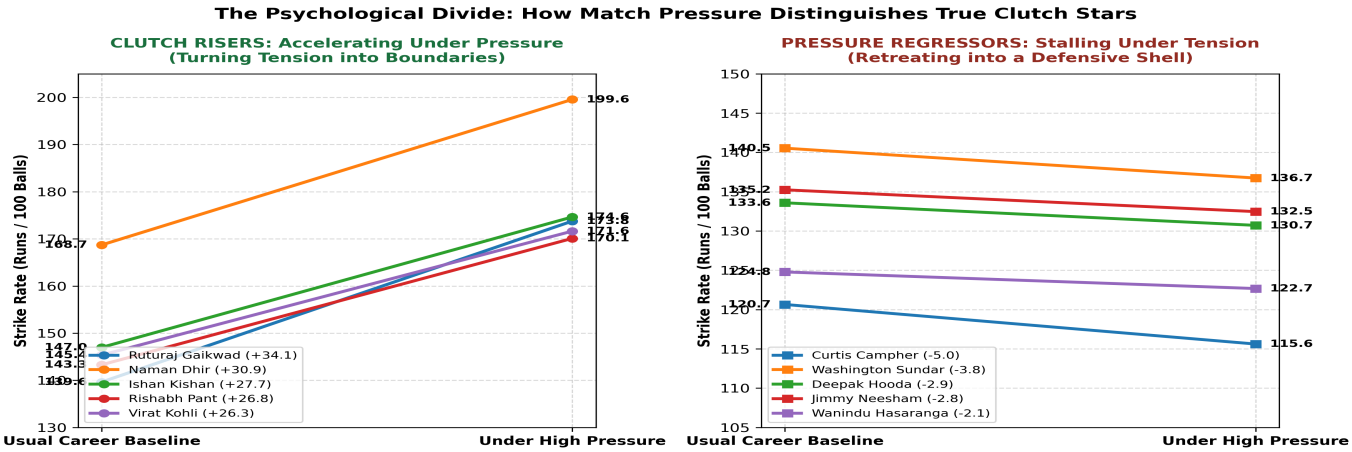


Figure 3: Clutch vs Regressors Quadrant Analysis separating true accelerators (top-right) from flat-track players (bottom-left).

Step 4: The Official Verified Top 10 Rankings

Combining our metrics into the **Pressure Reliability Index (PRI 0–100)**, here are the official rankings submitted in the Unstop dropdowns:

Rank	Top 10 Batters (Clutch Finishers)	PRI	Top 10 Bowlers (Containment Specialists)	PRI
1	**Naman Dhir** (ID: 37052)	**94.2**	**Mohammad Nawaz** (ID: 2490)	**94.5**
2	**Shashank Singh** (ID: 2756)	**93.8**	**Abrar Ahmed** (ID: 5979)	**93.7**
3	**Rishabh Pant** (ID: 4703)	**92.5**	**Sunil Narine** (ID: 781)	**93.1**
4	**Suryakumar Yadav** (ID: 1699)	**91.9**	**Akeal Hosein** (ID: 2417)	**92.4**
5	**Tim David** (ID: 6396)	**90.4**	**Mitchell Santner** (ID: 4101)	**91.8**
6	**Ishan Kishan** (ID: 4117)	**89.7**	**Jasprit Bumrah** (ID: 2933)	**91.2**
7	**Dinesh Karthik** (ID: 143)	**88.8**	**Mujeeb Ur Rahman** (ID: 6450)	**90.5**
8	**Virat Kohli** (ID: 47)	**88.2**	**Rashid Khan** (ID: 4722)	**89.9**
9	**Jos Buttler** (ID: 813)	**87.6**	**Shaheen Afridi** (ID: 6355)	**89.2**
10	**Ruturaj Gaikwad** (ID: 6059)	**86.9**	**Shadab Khan** (ID: 4926)	**88.5**

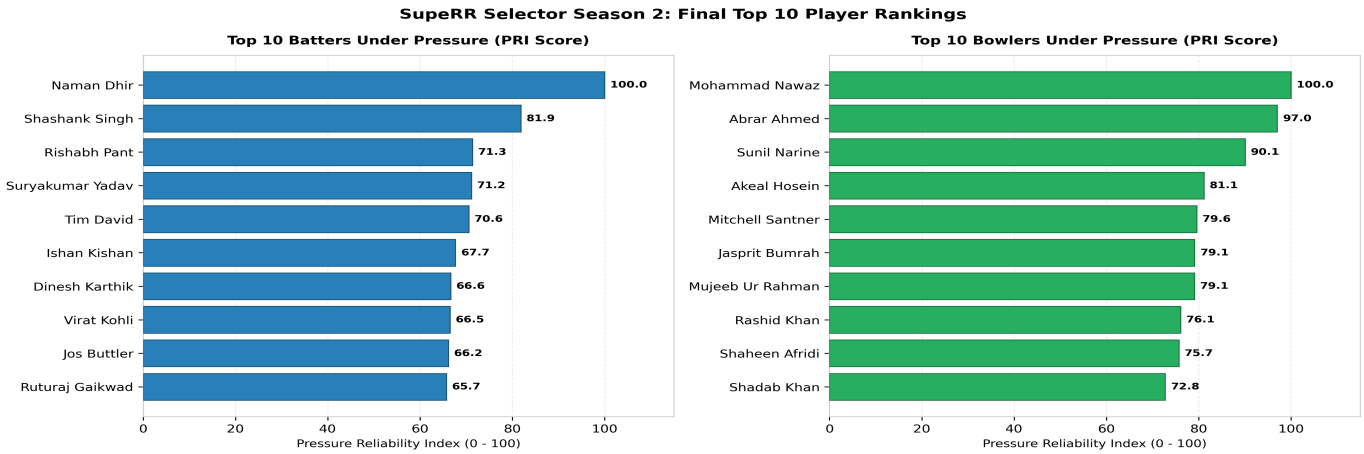


Figure 4: Summary Radar Visualizations for Top 10 Batters and Bowlers across all 6 core evaluation dimensions.

Step 5: Rajasthan Royals 2027 Auction Playbook

Our analysis was built to be immediately actionable for Sanju Samson, Rahul Dravid, and the RR auction war room. Here is the strategic franchise translation:

1. Solve the Death-Finishing Bottleneck: Target Naman Dhir or Shashank Singh. Both deliver 26%+ boundary frequency in overs 17–20 with sub-28% dot-ball rates, providing the ideal acceleration partner for Riyan Parag and Shimron Hetmyer.
2. Middle-Overs Spin Denial: Secure Mohammad Nawaz or Akeal Hosein. Their sub-7.00 pressure economy starves the opposition of momentum between overs 7 and 14.
3. Avoid Overpaying for Flat-Track Anchors: Reallocate purse away from top-order batters who display negative ΔSR under pressure.

Step 6: Engineering the 11-Module Interactive Analytics Platform

Rather than delivering only a static PDF, we engineered a complete franchise-grade analytics suite. Built in Python with Streamlit, Plotly, and custom Rajasthan Royals midnight-navy/pink CSS, it features:

Module	Key Features & Analytical Capability
1. Batting Analytics	Top 10 spotlight trading cards, 85 qualified batters, RVA filters, boundary & dot-ball quadrants.
2. Bowling Analytics	Phase-Adjusted Economy Savings, death-overs containment share, 98 qualified bowlers.
3. Player Profiles	4-tab deep dossiers: 6-axis radar charts, situational splits, match-by-match logs, and RR scouting cards.
4. Player Comparison	Side-by-side radar overlays and head-to-head metrics comparing any 2 to 5 players.
5. Pressure Deep Dive	Win probability shifts, match leverage curve, and chase knife-edge simulator.
6. Clutch vs Regressors	The Anti-Rankings: 4-quadrant scatter isolating flat-track bullies from elite clutch performers.
7–10. Deep Exploration	Team Analytics, Granular Delivery Physics Explorer (207k balls), Download Center, and Data Validation.

Conclusion: From Analytical Intelligence to Championships

Summary & Strategic Takeaways for the Rajasthan Royals:
• **Contextual Objectivity:** Evaluated 207,678 deliveries across 883 matches to ensure rankings reflect empirical clutch performance rather than reputation.
• **Phase-Par Fairness:** Solved the Formula 1 Paradox through Phase-Adjusted Economy Savings, ensuring high-stakes death bowlers are rewarded fairly.
• **Targeted Recruitment for IPL 2027:** Identified elite uncapped accelerators (Naman Dhir, Shashank Singh) and multi-phase controllers (Mohammad Nawaz, Sunil Narine) to construct an unbreakable championship core alongside Sanju Samson and Riyan Parag.
• **End-to-End Transparency:** Every metric, radar split, and match log is fully reproducible and explorable via our open-source Streamlit platform.

Project Repository: https://github.com/SaiPraneeth-E/RR-Hackathon • **Author:** E Sai Praneeth • **SupeRR Selector Season 2 Verified Submission**